

```

1  `timescale 1ns / 1ps
2  module AutoPlay(
3      input CLK,
4      input BTN,
5      input BTN_s,
6      output PWM
7  );
8      wire f_1sec;
9      reg [26:0] cnt_1sec=0;
10     reg [26:0] cnt_1sec_cmp=0;
11     reg [3:0] state_tone=0;
12     always @ (posedge CLK) begin
13         case(state_tone)
14             0:if(BTN) begin state_tone <= 1;
15                 if(BTN_s)
16                     cnt_1sec_cmp<=125-1;
17                 else
18                     cnt_1sec_cmp<=125000000-1;
19             end
20             1:if(f_1sec) state_tone <= 2;
21             2:if(f_1sec) state_tone <= 3;
22             3:if(f_1sec) state_tone <= 4;
23             4:if(f_1sec) state_tone <= 5;
24             5:if(f_1sec) state_tone <= 6;
25             6:if(f_1sec) state_tone <= 7;
26             7:if(f_1sec) state_tone <= 8;
27             8:if(f_1sec) state_tone <= 0;
28             default: state_tone <= 0;
29         endcase
30     end
31
32     always @(posedge CLK) begin
33         cnt_1sec <= cnt_1sec +1;
34         if(cnt_1sec ==cnt_1sec_cmp)
35             cnt_1sec <=0;
36         else if(state_tone==0)
37             cnt_1sec <=0;
38     end
39     assign f_1sec=(cnt_1sec==cnt_1sec_cmp);
40
41     reg [7:0] w_BTN;
42     MelodyScale MelodyScale_0(CLK, w_BTN, PWM);
43     always @(state_tone) begin
44         case(state_tone)
45             0:w_BTN <= 0;
46             1:w_BTN <= 1;
47             2:w_BTN <= 2;
48             3:w_BTN <= 4;
49             4:w_BTN <= 8;
50             5:w_BTN <= 16;
51             6:w_BTN <= 32;
52             7:w_BTN <= 64;
53             8:w_BTN <= 128;
54             default:w_BTN <= 0;
55         endcase
56     end
57 endmodule

```